

Endpoint rigidity and a superlinear lower bound for every die

Research note; endpoint theorem independently reviewed

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For each m , let $p_1, \dots, p_m$ be nondecreasing functions on a finite ordered probability space $(\{1, \dots, K\}, w)$, with positive row weights, and suppose

$$\mathbb{E}_w \prod_{j \in S} p_j = \frac{1}{|S| + 1} \quad (S \subseteq [m]). \quad (1)$$

All p_j take values in $[0, 1]$. The number of rows and their weights may depend on m . Put $A(q) = \prod_j p_j(q)$, and define the probability measure

$$\eta(q) = (m + 1)w_q A(q).$$

Rows with $A(q) = 0$ have zero η -mass. On all other rows define

$$r_j(q) = \frac{1 - p_j(q)}{p_j(q)}, \quad \lambda(q) = \sum_{j=1}^m r_j(q).$$

Theorem 1. *As $m \rightarrow \infty$, the distribution of λ under η converges weakly to the exponential distribution of mean one. This conclusion holds for every sequence of monotone representations satisfying (1), with no bound on K and no equal-weight assumption. Moreover,*

$$\lambda(K) \longrightarrow 0, \quad \prod_j p_j(K) \longrightarrow 1, \quad mw_K \longrightarrow 0.$$

Proof. For $0 \leq z < 1$, set

$$B_q(z) = \prod_j (p_j(q) + z(1 - p_j(q))).$$

The mixed identities, or equivalently the uniform law of the number of failures in the comparison tensor, give the polynomial identity

$$(m + 1)\mathbb{E}_w B_q(z) = \sum_{k=0}^m z^k. \quad (2)$$

For every fixed column j , the same identities give

$$(m + 1)\mathbb{E}_w \left[(1 - p_j) \prod_{i \neq j} (p_i + z(1 - p_i)) \right] = \frac{1}{m} \sum_{k=0}^{m-1} (k + 1)z^k \leq \frac{1}{m(1 - z)^2}. \quad (3)$$

Indeed, a particular failure set of size $k + 1$ has probability $1/((m + 1)\binom{m}{k+1})$, and there are $\binom{m-1}{k}$ such sets containing j .

Temporarily assign odds $+\infty$ to a zero entry. For $\varepsilon > 0$ let $D_\varepsilon = \{q : \max_j r_j(q) \geq \varepsilon\}$. This is an initial segment of the ordered rows. If it is nonempty, choose its last row q_* and

a column j_* with $r_{j_*}(q_*) \geq \varepsilon$. Monotonicity implies throughout D_ε that $p_{j_*} \leq (1 - p_{j_*})/\varepsilon$. Consequently,

$$(m+1)\mathbb{E}_w[B_q(z)\mathbf{1}_{D_\varepsilon}] \leq \frac{\varepsilon^{-1} + z}{m(1-z)^2}. \quad (4)$$

This argument includes rows containing zeros and does not cancel any zero factor. At $z = 0$ it also gives $\eta(D_\varepsilon) \leq 1/(m\varepsilon)$.

Take $\varepsilon_m = m^{-1/2}$ and let G_m be the complementary set of good rows. On G_m all entries are positive and $0 \leq r_j < \varepsilon_m$. Define

$$F_m(z) = (m+1)\mathbb{E}_w[B_q(z)\mathbf{1}_{G_m}].$$

Equations (2) and (4) show that $F_m(z) \rightarrow (1-z)^{-1}$ for each fixed $0 \leq z < 1$, and the same holds if z is replaced by a sequence tending to such a fixed value. For fixed $0 < z < 1$ put $z'_m = (1 + \varepsilon_m)z$. For all sufficiently large m , $z'_m < 1$ and

$$\log(1 + z'_m r) \geq z'_m r - \frac{(z'_m r)^2}{2} \geq zr \quad (0 \leq r \leq \varepsilon_m).$$

Also $\log(1 + zr) \leq zr$. Hence

$$F_m(z) \leq \mathbb{E}_\eta[e^{z\lambda}\mathbf{1}_{G_m}] \leq F_m(z'_m) \longrightarrow \frac{1}{1-z}. \quad (5)$$

The restricted measures have total mass tending to one, by the $z = 0$ case of (4). Their moment generating functions therefore converge on every compact subinterval of $(0,1)$ to $(1-z)^{-1}$. For completeness, the uniform bound in (5) at any $z_0 > 0$ gives tightness. A bound at $z_1 > z_0$ makes $e^{z_0\lambda}$ uniformly integrable, so every subsequential weak limit has moment generating function $(1-z)^{-1}$ on $(0,1)$. Uniqueness of moment generating functions identifies that limit as $\text{Exp}(1)$. The omitted η -mass tends to zero, proving the first assertion.

The last row has $A(K) > 0$. Odds decrease with the row, so $\lambda(K)$ is the least value on the support of η . Weak convergence to a law having positive mass in every interval $(0, a)$ implies $\lambda(K) \rightarrow 0$. The elementary inequality

$$e^{-\lambda(q)} \leq A(q) \leq 1$$

then gives $A(K) \rightarrow 1$. Finally, weak convergence to a continuous law implies that the maximal atom mass tends to zero; in particular $\eta(K) = (m+1)w_K A(K) \rightarrow 0$. Thus $mw_K \rightarrow 0$. $\square$

Proposition 1 (Uniform control of normalized moments). *Let $G_\varepsilon = \{q : \max_j r_j(q) < \varepsilon\}$ and put*

$$M_k = \mathbb{E}_\eta \left[\frac{\lambda^k}{k!} \mathbf{1}_{G_\varepsilon} \right].$$

For $0 \leq k \leq m$ and $k\varepsilon \leq 1$,

$$1 - \frac{(k+1)/\varepsilon + k}{m} \leq M_k \leq 1 + k\varepsilon.$$

In particular, with $\varepsilon = m^{-1/2}$,

$$|M_k - 1| \leq \frac{2k+1}{\sqrt{m}} \quad (0 \leq k \leq \lfloor \sqrt{m} \rfloor).$$

Proof. Write $e_k(r)$ for the elementary symmetric polynomial of degree k in the odds, with $e_0 = 1$, and put $a_k = \mathbb{E}_\eta[e_k(r)\mathbf{1}_{G_\varepsilon}]$. The proof of (4) is coefficientwise: on the bad initial segment,

$$B_q(z) \preceq (\varepsilon^{-1} + z)(1 - p_{j_*}) \prod_{i \neq j_*} (p_i + z(1 - p_i)),$$

where $\preceq$ denotes comparison of every coefficient. Equations (2)–(3) therefore give

$$1 - \frac{(k+1)/\varepsilon + k}{m} \leq a_k \leq 1.$$

At $k = m$ the term from coefficient m on the right derivative polynomial is absent, so the displayed lower bound is still valid.

The expansion of λ^k sums all ordered k -tuples of indices, whereas $k!e_k$ keeps only tuples with distinct indices. A union bound over pairs of coinciding positions gives, for $k \geq 2$,

$$0 \leq \lambda^k - k!e_k \leq \binom{k}{2} \left(\sum_i r_i^2 \right) \lambda^{k-2} \leq \binom{k}{2} \varepsilon \lambda^{k-1} \quad \text{on } G_\varepsilon.$$

Consequently

$$0 \leq M_k - a_k \leq \frac{(k-1)\varepsilon}{2} M_{k-1}.$$

Since $M_0 \leq 1$ and $M_1 = a_1 \leq 1$, induction gives $M_k \leq 1 + (k-1)\varepsilon \leq 1 + k\varepsilon$ whenever $k\varepsilon \leq 1$. The lower bound follows from $M_k \geq a_k$. Substitution of $\varepsilon = m^{-1/2}$ proves the final estimate. $\square$

Corollary 1. *Every sequence of equal-weight monotone representations (1) satisfies*

$$\frac{K}{m} \longrightarrow \infty.$$

Consequently, if $g(n)$ is the minimum number of faces on one die in any permutation-fair collection of n finite equiprobable dice, then

$$\frac{g(n)}{n} \longrightarrow \infty.$$

Together with the separately proved construction $g(n) = O(n^2)$, this places the exceptional-die optimum strictly above linear growth and at most at quadratic growth.

Proof. For equal row weights $w_K = 1/K$, Theorem 1 gives $m/K \rightarrow 0$. Comparison columns of any distinguished equiprobable die are monotone and satisfy (1). The argument applies to arbitrary sequences, so it gives the asserted universal asymptotic bound, not merely a bound for one selected family. $\square$

No rate of divergence is asserted here. The proof uses monotonicity only in the initial-segment estimate (4); positivity of individual product-mixture components alone does not give that estimate. The same conclusions hold at the opposite endpoint by replacing each $p_j(q)$ by $1 - p_j(K + 1 - q)$.